

39th Nordic Mathematical Contest

Solutions

Problems

1. Let n be a positive integer greater than 2. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying

$$\left(f(x+y)\right)^n = f(x^n) + f(y^n), \quad \text{for all integers } x, y.$$

2. Let p be a prime and suppose $2^{2p} \equiv 1 \pmod{2p+1}$. Prove that $2p+1$ is prime.¹
3. Let ABC be an acute triangle with orthocenter H and circumcenter O . Let E and F be points on the line segments AC and AB respectively such that $AEHF$ is a parallelogram. Prove that $|OE| = |OF|$.
4. Denote by S_n the set of all permutations of the set $\{1, 2, \dots, n\}$. Let $\sigma \in S_n$ be a permutation. We define the *displacement* of σ to be the number $d(\sigma) = \sum_{i=1}^n |\sigma(i) - i|$. We say that σ is *maximally displacing* if $d(\sigma)$ is the largest possible, i.e. if $d(\sigma) \geq d(\pi)$, for all $\pi \in S_n$.
- a) Suppose σ is a maximally displacing permutation of $\{1, 2, \dots, 2024\}$. Prove that $\sigma(i) \neq i$, for all $i \in \{1, 2, \dots, 2024\}$.
- b) Does the statement of part a) hold for permutations of $\{1, 2, \dots, 2025\}$?

Solutions

As usual, there are plenty of different kinds of solutions to each of the problems. Only for Problems 2 and 4, we present variants.

Problem 1

Substituting x for y in the functional equation one gets

$$f(2x)^n = f(x+x)^n = f(x^n) + f(x^n) = 2f(x^n),$$

for all $x \in \mathbb{Z}$. In particular, we have $f(0)^n = f(2 \cdot 0)^n = 2f(0^n) = 2f(0)$. If it were that $f(0) \neq 0$, we would have $f(x)^{n-1} = 2$, which is impossible, since $n-1 \geq 2$ and 2 is a prime. Hence, $f(0) = 0$, and by another substitution ($y = 0$) into the functional equation we get

$$f(x)^n = f(x+0)^n = f(x^n) + f(0^n) = f(x^n) + f(0) = f(x^n) + 0 = f(x^n),$$

¹This is a special case of Pocklington's theorem. A proof of this special case is required.

for $x \in \mathbb{Z}$. Combining our identities in displayed formulas, we conclude that

$$f(2x)^n = 2f(x^n) = 2f(x)^n.$$

Obviously zero function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ is a solution of the functional equation, as $0^n = 0 + 0$. We now claim that for every solution f of the functional equation, we have that $f(x) = 0$ holds for all $x \in \mathbb{Z}$. Let $x \in \mathbb{Z}$ be arbitrary. Suppose towards contradiction that $f(x) \neq 0$. Then also $f(2x) \neq 0$, and we may write

$$f(x) = 2^\alpha s \text{ and } f(2x) = 2^\beta t$$

where $\alpha, \beta \in \mathbb{N}$ and $s, t \neq 0$ are odd. Now we have that

$$2^{n\beta} t^n = f(2x)^n = 2f(x)^n = 2(2^\alpha s)^n = 2^{n\alpha+1} s^n.$$

However, as $n\beta = n\alpha + 1$ cannot hold (recall that $n > 2$), this is in contradiction with the fundamental theorem of arithmetic.

Problem 2

Solution with Euler's theorem

By Euler's theorem $2^{\varphi(2p+1)} \equiv 1 \pmod{2p+1}$. Hence, $2^{\gcd(\varphi(2p+1), 2p)} \equiv 1 \pmod{2p+1}$. Now $\gcd(\varphi(2p+1), 2p) = 1, 2, p$ or $2p$.

It is impossible that $2 = 2^1 \equiv 1 \pmod{2p+1}$ or $2^2 = 4 \equiv 1 \pmod{2p+1}$.

If $\gcd(\varphi(2p+1), 2p) = p$ or $2p$, then there is a prime $q \mid 2p+1$ such that $q \equiv 1 \pmod{p}$. Now $q \geq p+1$. Then $2q > 2p+1$. Hence, $q = 2p+1$, which proves the theorem.

Solution with Fermat's little theorem

Assume $q \mid 2p+1$ and q is a prime. By Fermat's little theorem $2^{q-1} \equiv 1 \pmod{q}$. Now $2^{\gcd(q-1, 2p)} \equiv 1 \pmod{q}$.

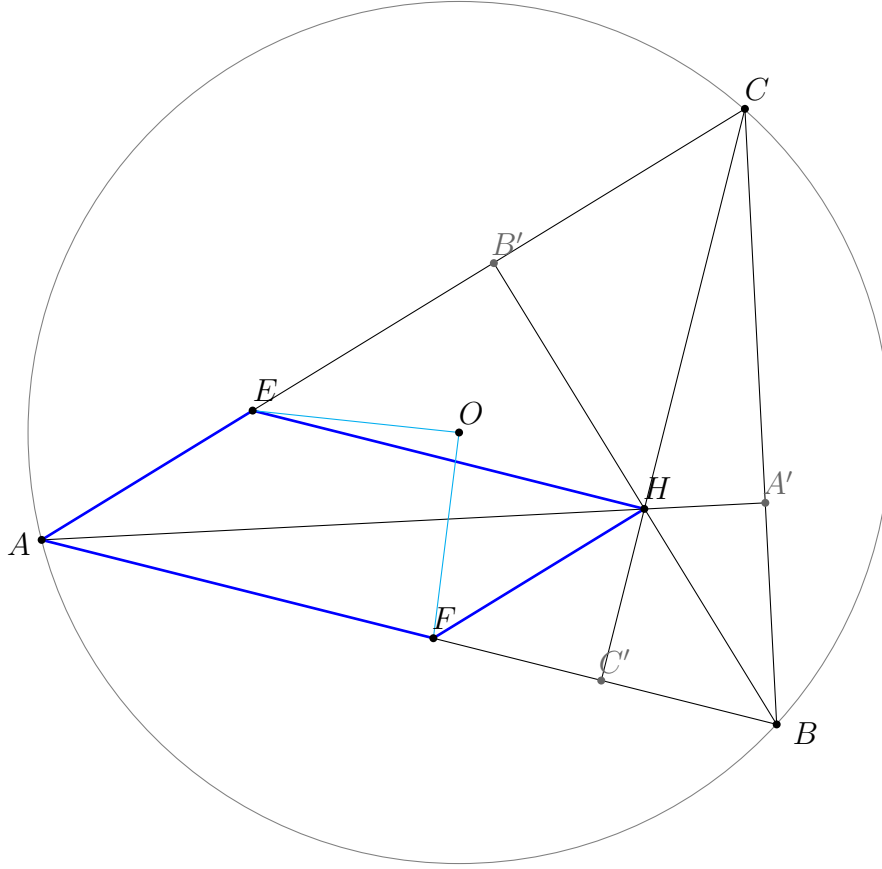
It is impossible that $2 = 2^1 \equiv 1 \pmod{q}$.

Now if $\gcd(q-1, 2p) = p$ or $2p$, then similarly as earlier, $q = 2p+1$.

Now $2^2 = 4 \equiv 1 \pmod{q}$ is only possible if $q = 3$. Hence, the only case remaining is the one with $2p+1 = 3^m$ for some positive integer m . It is impossible that $m = 1$. If $m \geq 2$, then we must have $2^{2p} \equiv 1 \pmod{9}$. However, if $2^\ell \equiv 1 \pmod{9}$, then $6 \mid \ell$. Hence, $3 \mid p$, in other words, then $p = 3$. In this case, $2p+1$ is not divisible by 9.

Problem 3

Let A' , B' and C' be the feet of the altitudes BC , AC and AB in $\triangle ABC$, respectively.



$AEHF$ being a parallelogram means that $AE \parallel FH$ and $EH \parallel AF$. As B' and E lie on AC , and C' and F on AB , this implies $AB' \parallel FH$ and $AC' \parallel EH$. Since H is an orthocentre, $AB'B$ and $AC'C$ are right triangles that share one angle, whence they are similar. As H lies on BB' , F on AC and $AB' \parallel FH$, we have $\triangle ABB' \sim \triangle FBH$, similarly $\triangle AC'C \sim \triangle EHC$. Hence, triangles HBF and HCE are similar, implying

$$|BF|/|FH| = |CE|/|EH| \Rightarrow |BF| \cdot |EH| = |CE| \cdot |FH| \Rightarrow |BF| \cdot |AF| = |CE| \cdot |AE|,$$

as $AEHF$ is a parallelogram. We conclude that the powers of points E and F with respect to the circumcircle of triangle ABC are the same, so they have the same distance from the circumcentre O , i.e., $|OE| = |OF|$.

Problem 4

The concepts of the problem generalize: Let $X \subseteq \mathbb{Z}$ be finite and σ a permutation on X , i.e., a bijection $X \rightarrow X$. Define the *displacement* of σ to be the number $d(\sigma) = \sum_{i \in X} |\sigma(i) - i|$. The permutation σ is called *maximally displacing* if $d(\sigma)$ is the largest possible, i.e. if $d(\sigma) \geq d(\pi)$, for all permutations π on X .

For a permutation σ on $X \subseteq \mathbb{Z}$, we also call $a \in X$ σ -forward if $\sigma(a) > a$, and σ -backward if $\sigma(a) < a$.

- a) Let σ be a permutation on a finite $X \subseteq \mathbb{Z}$. Suppose σ has distinct fixed points a and b . Consider $\rho = (a \ b) \circ \sigma$; so ρ maps elements otherwise as σ , but $\rho(a) = b$ and $\rho(b) = a$. Then

$$d(\rho) = \sum_{i \in X} |\rho(i) - i| = \sum_{i \in X \setminus \{a, b\}} |\rho(i) - i| + 2|b - a| = d(\sigma) + 2|b - a| > d(\sigma),$$

so σ is not maximally displacing.

Assume now X has an even size and σ has exactly one fixed point, c . Consider the cycle representation of σ . Then σ must have an odd cycle (of size at least 3). In that odd cycle, σ -forward and σ -backward elements cannot alternate. Hence, we may assume that there are $a, b \in X$ such that a and b are both σ -forward and $b = \sigma(a)$ (the backward case is handled similarly). Note that $a, b \neq c$. Consider $\rho = \sigma \circ (a \ b)$. Then

$$d(\rho) - d(\sigma) = |\rho(a) - a| + |\rho(b) - b| - |\sigma(a) - a| - |\sigma(b) - b| = (\sigma(b) - a) + 0 - (b - a) - (\sigma(b) - b) = 0.$$

So $d(\rho) = d(\sigma)$, but ρ has fixed points b and c , so neither of σ and ρ can be maximally displacing.

The claim of the item a now follows from what has been proved and that $\{1, 2, \dots, 2024\}$ has even size.

- b) We shall show that the result of item a does not hold, i.e., there is maximally displacing permutation on $\{1, 2, \dots, 2025\}$ with a fixed point.

Assume that finite $X \subseteq \mathbb{Z}$ has odd size with $|X| \geq 3$. Let σ be any maximally displacing permutation. Observe that such a permutation exists, as there are only finitely many permutations on X . As $|X|$ is odd, the cycle representation of σ has a cycle of odd size. We may assume that this cycle is not a singleton. Then by the argument used in item a, we see once again that there is ρ such that $d(\rho) = d(\sigma)$ and ρ has a fixed point b . In particular, there is a maximally displacing ρ on $\{1, 2, \dots, 2025\}$.

Auxiliary (constructive) solution to item b. Let π be a permutation on a finite $X \subseteq \mathbb{Z}$. Suppose X has at least two elements; denote by m the least, and by M the greatest element. We claim that there exists a permutation ρ on X with $d(\rho) \geq d(\pi)$ and $\rho(m) = M$, and such that $\rho(M) = \pi(M)$ unless $\pi(M) = M$. We may assume $\pi(m) \neq M$. Consider $\rho = (a \ M) \circ \pi$ where $a = \pi(m)$. Write also $b = \pi^{-1}(M)$. Then ρ and π differ only at m and b , so

$$\begin{aligned} d(\rho) - d(\pi) &= |M - m| + |a - b| - (|a - m| + |M - b|) \\ &= M - m + |a - b| - (a - m + M - b) \\ &= |a - b| - (a - b) \geq 0. \end{aligned}$$

Observe that now $\rho(m) = M \neq m$ and by a similar argument we may show that there is a permutation σ on X with $\sigma(M) = m$, $\sigma(m) = \rho(m) = M$ and $d(\sigma) \geq d(\rho)$. We conclude that there is a maximally displacing permutation σ swapping m and M .

Suppose further that $X \subseteq \mathbb{Z}$ is finite, but now that its size is odd (possibly one). Let $X = \{a_1, a_2, \dots, a_{2m+1}\}$ be the enumeration of its elements in increasing order. We are now ready for proving by induction that for every such X , the permutation

$$\sigma_X = (a_1 \ a_{2m+1})(a_2 \ a_{2m+2}) \cdots (a_m \ a_{m+2})$$

of the set X , e.g., the permutation that for distinct indexed $i, j \in \{1, \dots, 2m+1\}$ with $i+j = 2m+2$ swaps a_i and a_j and keeps a_{m+1} fixed, is indeed maximally displacing. For

$m = 0$ this is trivial, so assume $m > 0$. Suppose σ is maximally displacing permutation of X swapping $m = \min X \neq M = \max X$. Write $Y = X \setminus \{m, M\}$ and $\sigma^\dagger = \sigma \upharpoonright Y$. By induction assumption, we have $d(\sigma_Y) \geq d(\sigma^\dagger)$. But obviously

$$d(\sigma_X) - d(\sigma) = d(\sigma_Y) - d(\sigma^\dagger) \geq 0,$$

as $s_X(m) = M = \sigma(m)$ and $s_X(M) = m = \sigma(M)$. Hence, σ_X is maximally displacing as well.

In summary, there is σ_X is a maximally displacing permutation of the set $X = \{1, 2, \dots, 2025\}$ with $(1013) = 1013$, so the statement in item a does not hold in this case.